# • Regresión Logistica (Clasificación Binaria)

$y_i \in \{0, 1\}$

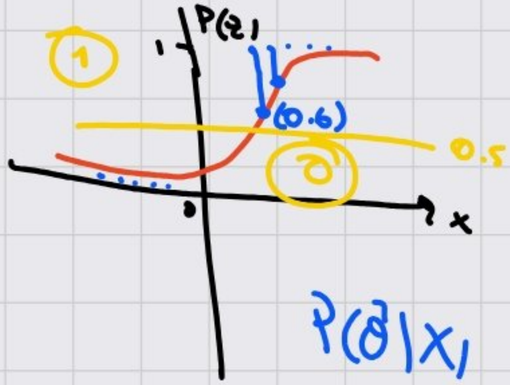
## > Modelo

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x_1 + \ldots + \beta_m x_m = X\beta$$

(params.)

log-odds

$$P = \frac{1}{1+e^{-X\beta}}$$




## > Función de Costo / Error

Función de Verosimilitud

$$L = \prod_{i=1}^{n} P_i^{y_i} (1-P_i)^{(1-y_i)}$$

BCE Binary Cross Entropy

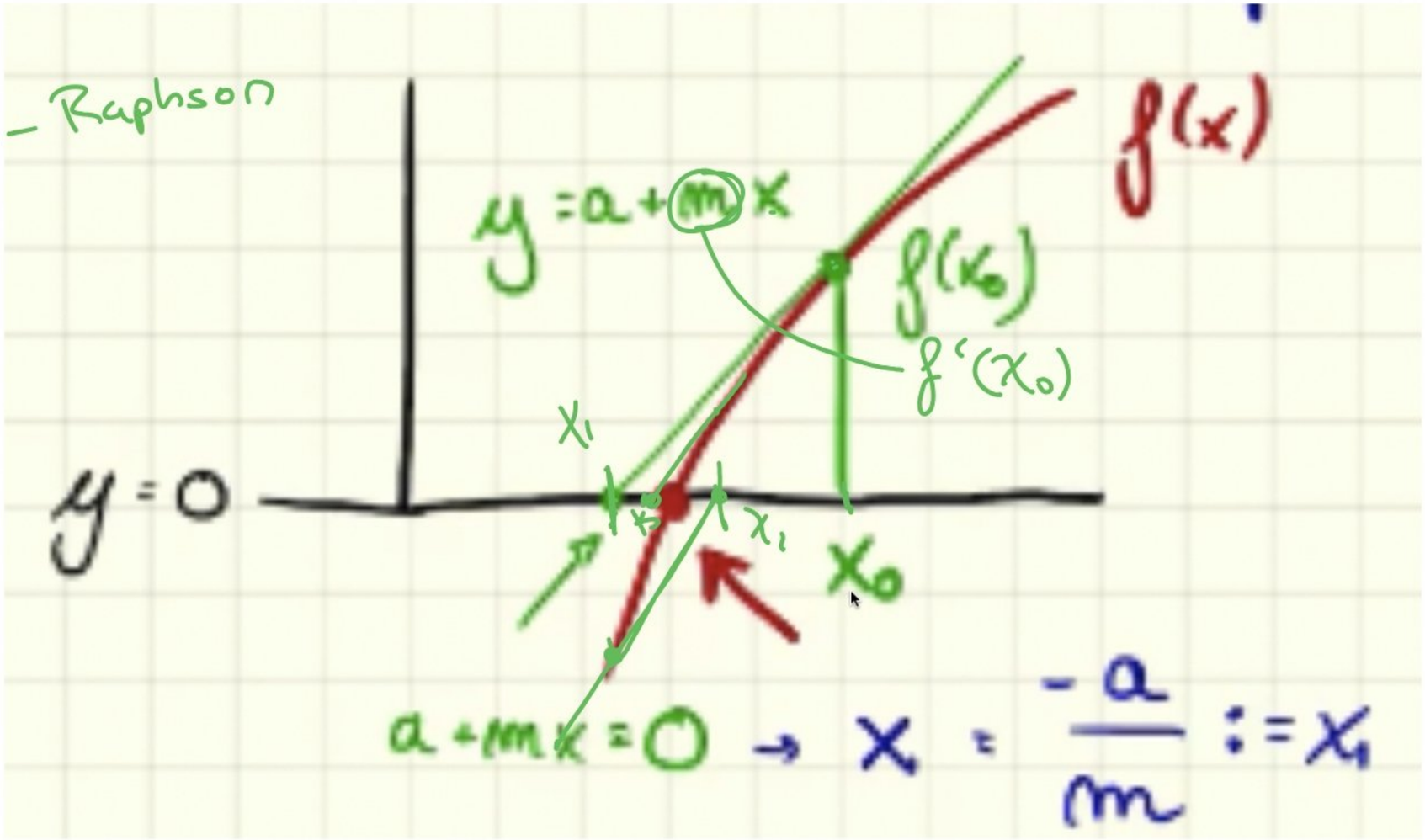
$$\ell = \ln(L) = \sum_{i=1}^{n} y_i \ln(P_i) + (1-y_i)\ln(1-P_i)$$

→ Optimización $\operatorname{argmax}(\ell)$

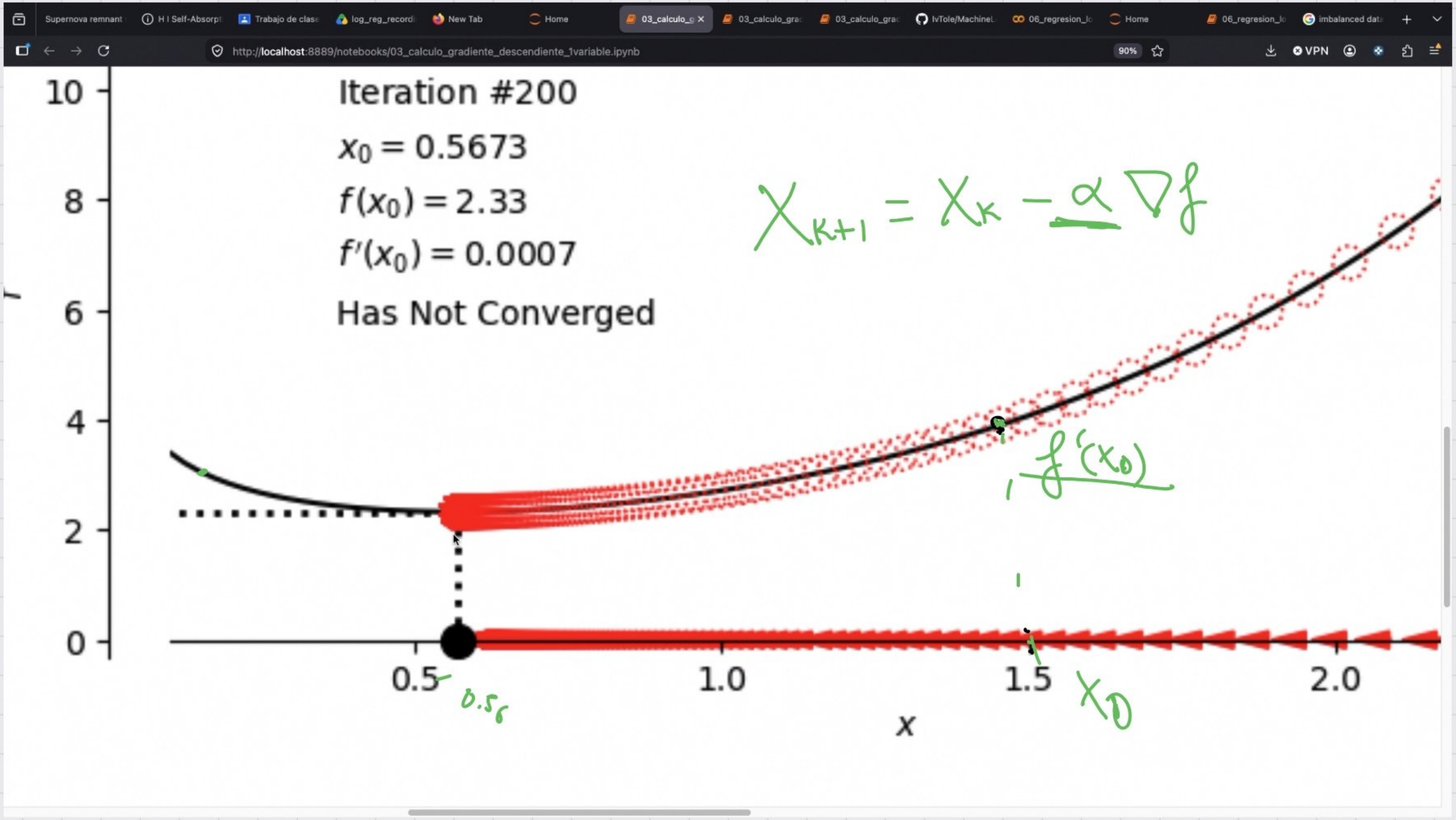
$$\frac{\partial \ell}{\partial \beta} = X(y-P) = 0 \; ; \quad \frac{\partial^2 \ell}{\partial \beta^2} = X \cdot W \cdot X^T$$

∂ No tiene sol. analítica → Método numérico B.V

Newton-Raphson




$$a + mx = 0 \rightarrow x = \frac{-a}{m} := x_1$$


Iteration #200
$x_0 = 0.5673$
$f(x_0) = 2.33$
$f'(x_0) = 0.0007$
Has Not Converged
$x_{k+1} = x_k - \alpha \nabla f$
$f'(x_0)$
$x_0$
0.56
10
8
6
4
2
0
0.5
1.0
1.5
2.0
x

Supernova remnant | H I Self-Absorpt | Trabajo de clase | log_reg_record | New Tab | Home | 03_calculo_g | 03_calculo_grad | 03_calculo_grad | IvTole/MachineL | 06_regresion_lo | Home | 06_regresion_lo | imbalanced data
http://localhost:8889/notebooks/03_calculo_gradiente_descendiente_1variable.ipynb 90% VPN

le es:

$$x_{k+1} = x_k - \alpha \cdot f'(x_k),$$

$$\frac{\partial \ell}{\partial \beta} = 0 = X(y - P); \quad \frac{\partial^2 \ell}{\partial \beta^2} = X \cdot W \cdot X^T$$

$y_i = \{0, 1\}$

$$P_i = \frac{1}{1 + e^{-X_i \beta}}$$

$$W = \begin{pmatrix} P_1(1-P_1) & 0 & 0 & \cdots \\ 0 & P_2(1-P_2) & 0 & \vdots \\ 0 & 0 & P_3(1-P_3) & \vdots \\ \cdots & \cdots & \cdots & \ddots \end{pmatrix}$$

0.25 0.25

$$W = \operatorname{diag}(P_i(1-P_i))$$

## Newton-Raphson

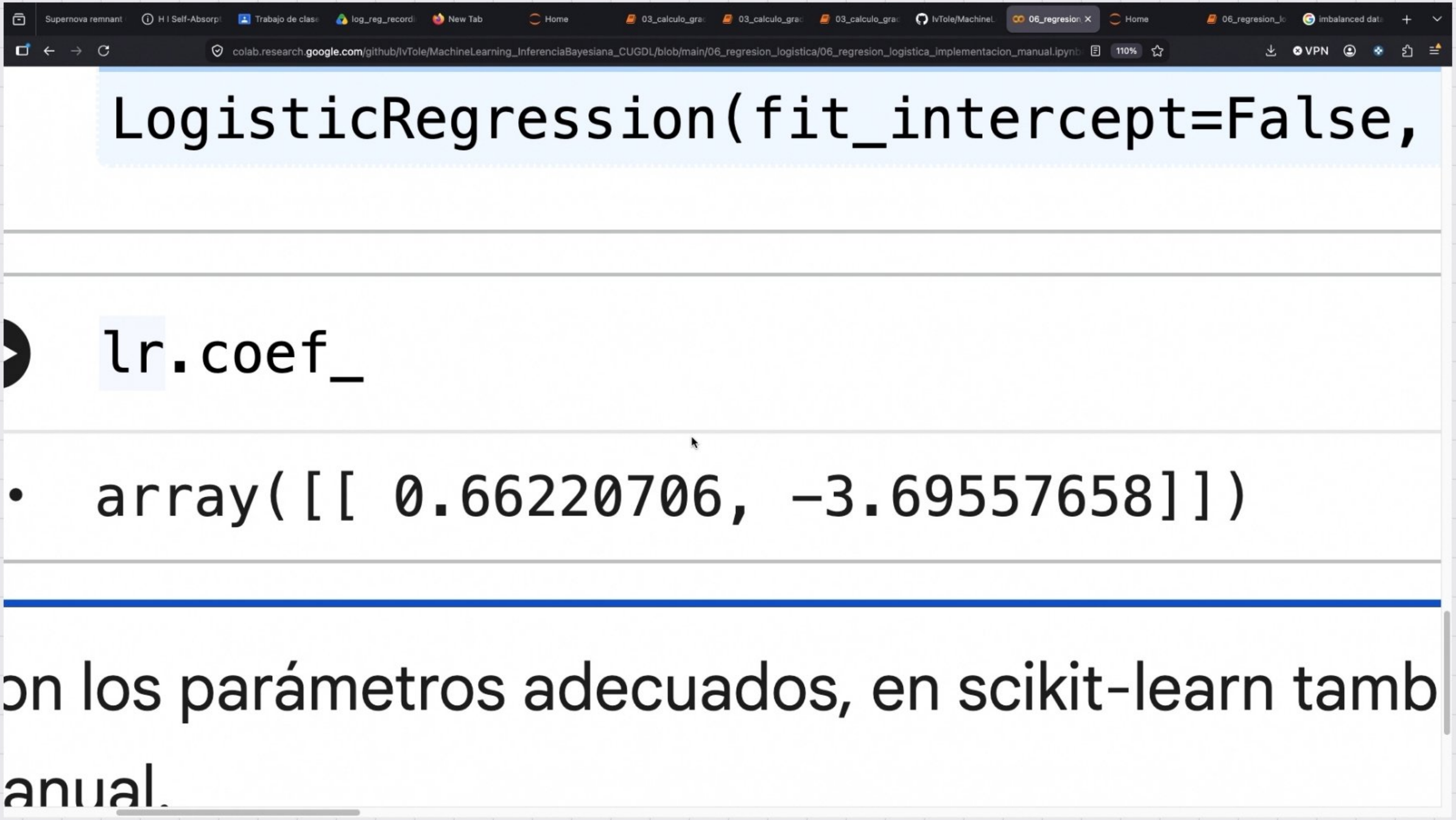
$$\beta_{k+1} = \beta_k - \frac{f(\beta_k)}{f'(\beta_k)} = \beta_k - (X(y-p))(X \cdot W \cdot X^T)^{-1}$$

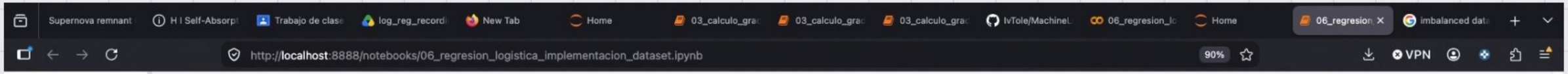
DATOS (1000)

800 : Clase 0 $P = 0.8$

200 : Clase 1 $1 - P = 0.2$

Balance


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imbalanced data
colab.research.google.com/github/IvTole/MachineLearning_InferenciaBayesiana_CUGDL/blob/main/06_regresion_logistica/06_regresion_logistica_implementacion_manual.ipynb
110%
VPN
LogisticRegression(fit_intercept=False,
lr.coef_
array([[ 0.66220706, -3.69557658]])
on los parámetros adecuados, en scikit-learn tamb
anual.




41]:
```
y
no     3668
yes     451
Name: count, dtype: int64
```

~90%.

~10%.

# Observaciones iniciales

- El dataset contiene aproximadamente 4119 fi
- La variable objetivo `y` es binaria: `yes` o `n`